

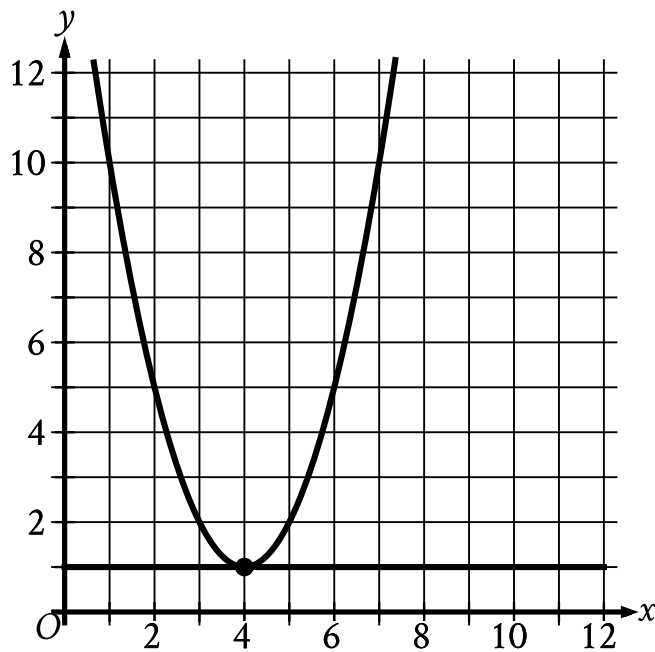
● EASY

● ADVANCED MATH

Advanced Math

09

10 questions · ~15 min



- For the parabola in the system:
 - The parabola opens upward.
 - The vertex is at point (4 comma 1).
 - The parabola passes through the following points:
 - (3 comma 2)
 - (4 comma 1)
 - (5 comma 2)
- For the line in the system:
 - The line is horizontal.
 - The line passes through the following points:
 - (0 comma 1)
 - (4 comma 1)

The graph of a system of a linear and a quadratic equation is shown. What is the solution x, y to this system?

A. 0,0

B. -4, 1

C. 4, -1

D. 4, 1

Q2

ADVANCED MATH

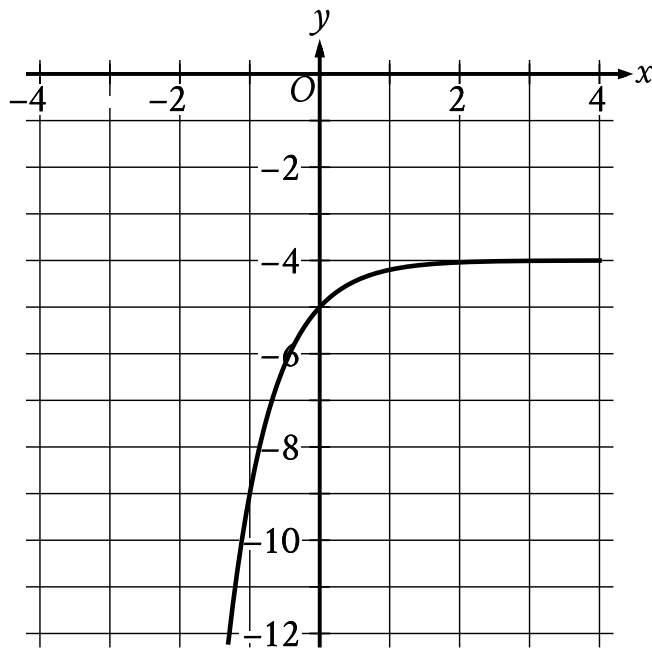
Which expression is equivalent to $5x^2 - 50xy^2$?

A. $5xx - 10y^2$

B. $5xx - 50y^2$

C. $5x^2 10xy^2$

D. $5x^2 50xy^2$



- Moving from left to right:
 - The curve passes from quadrant 3 to quadrant 4.
 - In quadrant 3, the curve trends up sharply to point (0 comma negative 5).
 - In quadrant 4, the curve trends up gradually.
- As x increases, the curve approaches the line y equals negative 4.
- The curve passes through the following points:
 - (negative 1 comma negative 9)
 - (0 comma negative 5)
 - (1 comma negative 4.2)

What is the y -intercept of the graph shown?

A. -1, -9

B. 0, -5

C. 0, - 4

D. 0, 0

Q4

ADVANCED MATH

$$\frac{x^2}{25} = 36$$

What is a solution to the given equation?

A. 6

B. 30

C. 450

D. 900

Q5

ADVANCED MATH

Which expression is equivalent to $8 + d^2 + 3$?

A. $d^2 + 24$

B. $d^2 + 11$

C. $d^2 + 5$

D. $d^2 - 11$

The function g is defined by $gx = x^2 + 9$. For which value of x is $gx = 25$?

- A. 4
- B. 5
- C. 9
- D. 13

Q7

GRID-IN

ADVANCED MATH

$$x - 2 = 9$$

What is one possible solution to the given equation?

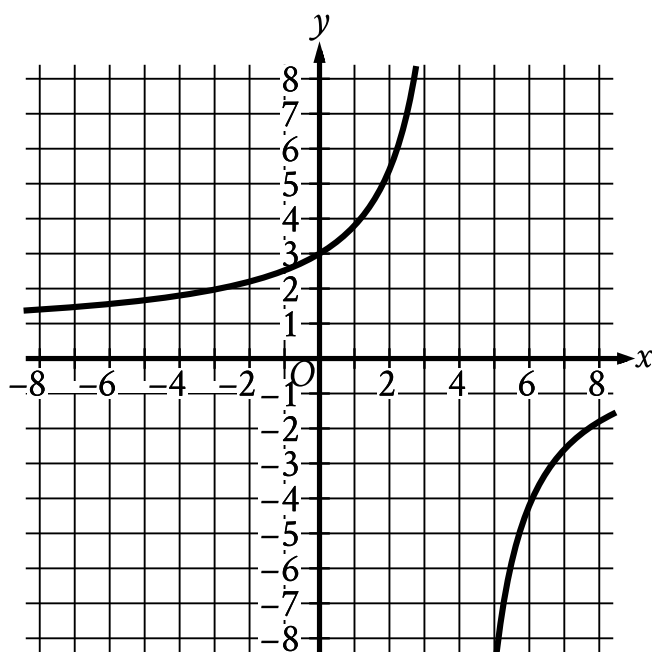
STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Which expression is equivalent to $9x^2 + 5x$?

- A. $x9x + 5$
- B. $5x9x + 1$
- C. $9xx + 5$
- D. $x^29x + 5$



- For the first curve:
 - Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up gradually to point $(0, 3)$.
 - In quadrant 1, the curve trends up sharply.
 - As x increases, the curve approaches the line x equals 4.
 - As x decreases, the curve approaches the line y equals 1.
 - The curve passes through the following points:
 - $(-3, 2)$
 - $(0, 3)$
- For the second curve:
 - Moving from left to right:
 - The curve is in quadrant 4.
 - In quadrant 4, the curve trends up sharply.
 - As x increases, the curve approaches the line y equals 1.

- As x decreases, the curve approaches the line x equals 4.
- The curve passes through the following point:
 - (6 comma negative 4)

The graph of $y = f(x)$ is shown in the xy -plane. What is the value of $f(0)$?

- A. -3
- B. 0
- C. $\frac{3}{5}$
- D. 3

Q10

ADVANCED MATH

$$p + 61 = 65$$

Which value is a solution to the given equation?

- A. $\frac{65}{61}$
- B. 4
- C. 126
- D. 130